

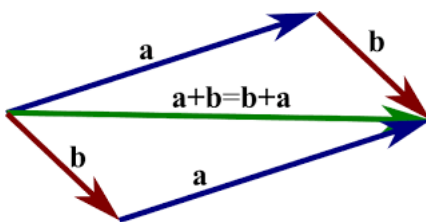
This document is a **transcript** of the lecture, with extra summary and vocabulary sections for your convenience. Due to time constraints, the notes may sometimes only contain limited illustrations, proofs, and examples; for a more thorough discussion of the course content, **consult the textbook**.

Summary

Quick summary of today's notes. Lecture starts on next page.

- A **vector** is a matrix with one column. We add and subtract vectors of the same size by doing the operations component-wise: $\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \pm \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} u_1 \pm v_1 \\ u_2 \pm v_2 \\ u_3 \pm v_3 \end{bmatrix}$ and $c \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} cu_1 \\ cu_2 \\ cu_3 \end{bmatrix}$ for $c \in \mathbb{R}$.
- Let n be a positive integer and define $\mathbb{R}^n$ to be the set of vectors with n rows.
- We reuse the symbol 0 to mean the vector in $\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^n$ whose entries are all zeros.
- Visualize vectors $a = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \in \mathbb{R}^2$ as arrows from the origin $(x, y) = (0, 0)$ to $(x, y) = (a_1, a_2)$.

The sum $a + b$ for $a, b \in \mathbb{R}^2$ is then the diagonal of the parallelogram with sides a and b :



- A **linear combination** of vectors $v_1, v_2, \dots, v_p \in \mathbb{R}^n$ is a vector of the form $c_1v_1 + c_2v_2 + \dots + c_pv_p$ where $c_1, c_2, \dots, c_p \in \mathbb{R}$ are numbers. The set of all linear combinations of $v_1, v_2, \dots, v_p \in \mathbb{R}^n$ is called the **span** of the vectors and is denoted by $\mathbb{R}\text{-span}\{v_1, v_2, \dots, v_p\}$.

Example: if $e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ and $e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ then $\mathbb{R}\text{-span}\{e_1, e_3\} = \left\{ \begin{bmatrix} a \\ 0 \\ b \\ 0 \end{bmatrix} : a, b \in \mathbb{R} \right\}$.

- If $x_1, x_2, \dots, x_n$ are variables and $a_1, a_2, \dots, a_n, b \in \mathbb{R}^m$ are vectors then we refer to

$$x_1a_1 + x_2a_2 + \dots + x_na_n = b$$

as a **vector equation**. It has the same solutions as the linear system with augmented matrix

$$\begin{bmatrix} a_1 & a_2 & a_3 & \dots & a_n & b \end{bmatrix}.$$

The vector $b \in \mathbb{R}^m$ is in the span of the vectors $a_1, a_2, \dots, a_n \in \mathbb{R}^m$ precisely when this linear system has a solution. (And we can figure out if this happens by computing the reduced echelon form of the system's augmented matrix and checking whether the last column contains a pivot.)

1 Last time: row reduction to (reduced) echelon form

The *leading entry* in a nonzero row of a matrix is the first nonzero entry from left going right. For example, the row $\begin{bmatrix} 0 & 0 & 7 & 0 & 5 \end{bmatrix}$ has leading entry 7, which occurs in the 3rd column.

Definition. A matrix with m rows and n columns is in *echelon form* if it has the following properties:

1. If a row is nonzero, then every row above it is also nonzero.
2. The leading entry in a nonzero row is in a column to the right of the leading entry in the row above.
3. If a row is nonzero, then every entry below its leading entry in the same column is zero.

In echelon form: $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 3 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. Not in echelon form: $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 3 & 5 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 3 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 2 & 0 & 0 \\ 1 & 3 & 5 & 0 \\ 0 & 0 & 4 & 5 \end{bmatrix}$.

Definition. A matrix is in *reduced echelon form* if

1. The matrix is in echelon form.
2. Each nonzero row has leading entry 1.
3. The leading 1 in each nonzero row is the only nonzero number in its column.

① rescaling rows ② swapping rows

Theorem. Each matrix A is row equivalent $\rightarrow$ means can be transformed by $\rightarrow$ or ③ doing row replacements.

We denote this matrix by $\text{RREF}(A)$.

The *row reduction algorithm* is a way of constructing $\text{RREF}(A)$ from A . This algorithm is something you should memorize and be able to perform quickly. The algorithm is illustrated by the following example:

Example. Writing $\rightarrow$ to indicate a sequence of row operations, we have

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 1 \\ 1 & 3 & 9 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & 2 & 8 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

and the last matrix is the reduced echelon form of the first matrix.

Consider the nonzero rows of $\text{RREF}(A)$. In these row find the first nonzero entry from left to right.

If one of these leading entries is in column j , then j is a *pivot column* of A . For example if

$$\text{RREF}(A) = \begin{bmatrix} 1 & -2 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

then the leading entries are in positions $(1, 1)$ and $(2, 3)$ so the pivot columns of A are 1 and 3.

If A is the augmented matrix of a linear system in variables $x_1, x_2, \dots, x_n$, then we say that x_i is a *basic variable* if i is a pivot column of A and that x_i is a *free variable* if i is not a pivot column of A .

To determine the basic and free variables of the system, we have to perform the row reduction algorithm to figure out what $\text{RREF}(A)$ is first. Once we have done this, we can conclude that:

- The system has 0 solutions if the last column is a pivot column of A .
- The system has ∞ solutions if the last column is not a pivot column but there is ≥ 1 free variable.
- The system has 1 solution if there are no free variables, and the last column is not a pivot column.

$\rightarrow$ only way for the last column to have a pivot is it
 $\text{RREF}(A) = \begin{bmatrix} \text{???} \\ 0 & 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & 0 & \dots & 0 & 0 \end{bmatrix}$ but then $\text{RREF}(A)$ corresponds to a linear system with the equation $0x_1 + 0x_2 + 0x_3 + \dots + 0x_n = 1$
 $0 = 1 \rightarrow \text{Always false.}$

Moreover, here's how you can solve the system: write down the equations in the linear system whose augmented matrix is $\text{RREF}(A)$. Each nontrivial equation starts with a basic variable x_i and has the form

$$x_i + (\text{an expression involving free variables}) = (\text{a number})$$

After moving the expression involving free variables to the right side of the equation we get

$$x_i = (\text{a number}) - (\text{an expression involving free variables}).$$

To form the general solution to our original linear system, we just choose arbitrary values for the free variables and express the basic variables using these equations.

Example. The linear system

$$\begin{cases} 3x_2 - 6x_3 = 6 \\ 3x_1 - 7x_2 + 8x_3 = -5 \\ 3x_1 - 9x_2 + 12x_3 = -9 \end{cases} \quad \text{has augmented matrix} \quad A = \begin{bmatrix} 0 & 3 & -6 & 6 \\ 3 & -7 & 8 & -5 \\ 3 & -9 & 12 & -9 \end{bmatrix}$$

Its reduced echelon form is

$$\text{RREF}(A) = \begin{bmatrix} 1 & 0 & -2 & 3 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

This means that the pivot columns of A are columns 1 and 2, so x_1 and x_2 are basic variables while x_3 is a free variable. The last column is not a pivot column, so the linear system has infinitely many solutions.

The linear system with augmented matrix $\text{RREF}(A)$ is

$$\begin{cases} x_1 - 2x_3 = 3 \\ x_2 - 2x_3 = 2 \\ 0 = 0 \end{cases} \quad \text{which we can rewrite as} \quad \begin{cases} x_1 = 3 + 2x_3 \\ x_2 = 2 + 2x_3 \\ 0 = 0. \end{cases}$$

We choose an arbitrary value for the free variable $x_3 = a \in \mathbb{R}$.

Then the general solution is $(x_1, x_2, x_3) = (3 + 2a, 2 + 2a, a)$ where a can be any number.

Corollary. Suppose a linear system with m equations and n variables has augmented matrix A .

If $\text{RREF}(A)$ has the form $\left[\begin{array}{ccc|c} 1 & & & b_1 \\ & 1 & & b_2 \\ & & \ddots & \vdots \\ & & & 1 & b_m \end{array} \right]$ where all blank entries are zero, then the linear system has exactly one solution, and this solution is given by $(x_1, x_2, \dots, x_m) = (b_1, b_2, \dots, b_m)$.

Proof. The starting linear system has the same solutions as the linear system whose augmented matrix is $\text{RREF}(A)$. But the second system consists of the equations $x_1 = b_1, x_2 = b_2, \dots, x_m = b_m$. $\square$

2 Vectors better notation $\Rightarrow$ start to think about the geometry of solutions to linear systems.

Until we discuss vector spaces, the term *vector* will always refer to a matrix with exactly one column:

$$\begin{bmatrix} 1 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 3 \\ -1 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 1 \\ 2 \\ 3 \\ 5 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} \sqrt{7} \\ \sqrt{6} \end{bmatrix}.$$

Sometimes people refer to vectors defined like this as *column vectors*.

We write a general vector as $v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ where each v_i is a real number.

Two vectors u and v are equal if they have the same number of rows and the same entries in each row.

The *size* of a vector is its number of rows. We can add two vectors of the same size:

Vector addition $\leftarrow$ *inputs vectors u and v with n rows.*

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} + \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}.$$

Output: vector $u+v$ also with n rows

Note that $u + v = v + u$. If u and v don't have the same size then $u + v$ is not defined.

If v is a vector and $c \in \mathbb{R}$ is a *scalar*, i.e., a real number, then we define

Scalar multiplication $\leftarrow$

$$cv = c \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} cv_1 \\ cv_2 \\ \vdots \\ cv_n \end{bmatrix}.$$

*$u+v$ } Not defined if u and v have different number of rows.
 $u-v$ }*

We call the new vector cv the *scalar multiple* of v by c .

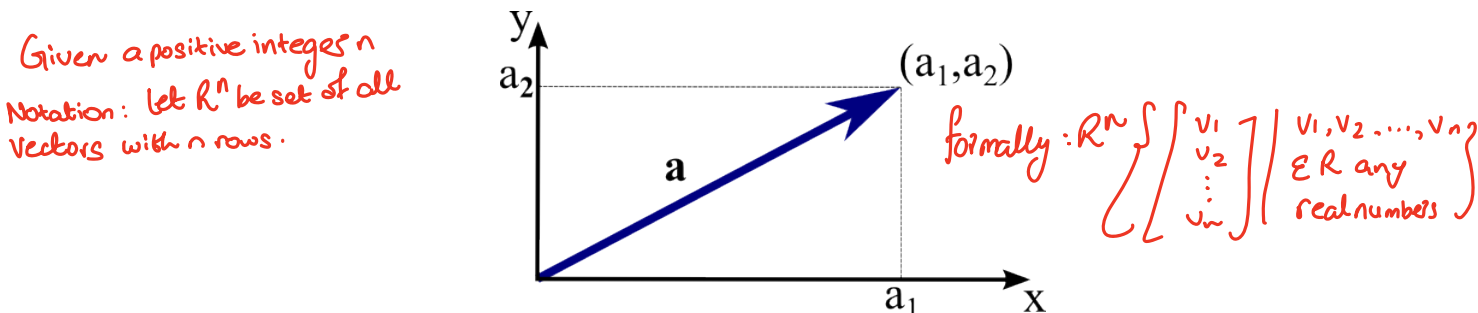
For example, we have

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \quad \text{and} \quad - \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

We define *subtraction* of vectors as addition after multiplying by the scalar -1 :

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix} - \begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} + (-1) \begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ -5 \end{bmatrix} = \begin{bmatrix} 0 \\ -7 \end{bmatrix}.$$

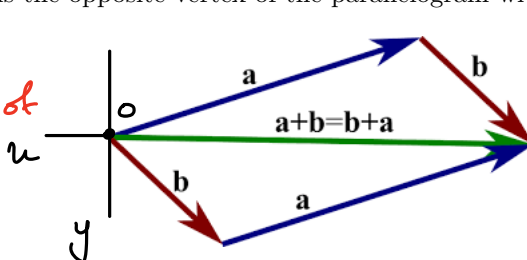
We write $\mathbb{R}^n$ for the set of all vectors with exactly n rows. Vectors $a = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \in \mathbb{R}^2$ can be identified with arrows in the Cartesian plane from the origin to the point $(x, y) = (a_1, a_2)$:



Proposition. The sum $a + b$ of two vectors $a, b \in \mathbb{R}^2$ is the vector represented by the arrow from the

origin to the point that is the opposite vertex of the parallelogram with sides a and b :

given vectors $\vec{a}$ and $\vec{b}$
the sum $\vec{a} + \vec{b}$ is the diagonal of
the $\Rightarrow$ they determine.



$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \end{bmatrix}$$

$$a_1 > 0, b_1 > 0 \\ a_2 > 0, b_2 < 0$$

Proof. We can write $\frac{a_2}{a_1} = \frac{(a_2+b_2)-b_2}{(a_1+b_1)-b_1}$ and $\frac{b_2}{b_1} = \frac{(a_2+b_2)-a_2}{(a_1+b_1)-a_1}$.

The fractions $\frac{a_2}{a_1}$ and $\frac{b_2}{b_1}$ are the slopes of the lines through the origin containing the vectors a and b .

The other two fractions are the slopes of the lines (1) between the endpoints of b and $a+b$ and (2) between the endpoints of a and $a+b$.

The first line of the proof shows that line (1) is parallel to a , and line (2) is parallel to b .

Therefore lines (1) and (2) are the other two sides of the unique parallelogram with sides a and b .

The endpoint of $a+b$ is where lines (1) and (2) intersect.

Therefore this endpoint is the vertex of the parallelogram opposite the origin. $\square$

The **zero vector** $\mathbb{R}^n$ is the vector $\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$ whose entries are all zero.

linear combination:
A new vector formed from given vectors in $\mathbb{R}^n$ from simultaneous addition and scalar multiplication.

We use the same symbol “0” to mean both the number zero and the zero vector in $\mathbb{R}^n$ for any n . You may have to use context to figure out which number or zero vector “0” means in a given expression.

We have $0 + v = v + 0 = v$ for any vector v .

Definition. Suppose $v_1, v_2, \dots, v_p \in \mathbb{R}^n$ are vectors and $c_1, c_2, \dots, c_p \in \mathbb{R}$ are **scalars**, i.e., numbers.

The vector $y = c_1v_1 + c_2v_2 + \dots + c_pv_p$ is called a **linear combination** of $v_1, v_2, \dots, v_p$.

We say that y is “the linear combination of $v_1, v_2, \dots, v_p$ with **coefficients** $c_1, c_2, \dots, c_p$.”

Example. Suppose $a = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$ and $c = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$. Is c a linear combination of a and b ?

If it were, we could find numbers $x_1, x_2 \in \mathbb{R}$ such that $x_1a + x_2b = c$, or equivalently such that

$$\begin{aligned} x_1 + 2x_2 &= 7 \\ -2x_1 + 5x_2 &= 4 \\ -5x_1 + 6x_2 &= -3. \end{aligned}$$

So to answer our question we need to determine if this linear system has a solution.

To do this, use row reduction:

$$A = \begin{bmatrix} 1 & 2 & 7 \\ -2 & 5 & 4 \\ -5 & 6 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 7 \\ 0 & 9 & 18 \\ 0 & 16 & 32 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \rightarrow \text{RREF}(A) = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns of A are 1 and 2: the last column is **not** a pivot column. Therefore our linear system is consistent, which means that c is a linear combination of a and b .

We generalize this example with the following statement.

Proposition. A vector equation of the form $x_1a_1 + x_2a_2 + \cdots + x_na_n = b$ where $x_1, x_2, \dots, x_n$ are variables and $a_1, a_2, \dots, a_n, b \in \mathbb{R}^m$ are vectors, has the same solutions as the linear system with augmented matrix

$$\begin{bmatrix} a_1 & a_2 & a_3 & \cdots & a_n & b \end{bmatrix}. \quad (**)$$

This notation means the matrix whose i th column is a_i and last column is b .

In other words, the vector b is a linear combination of $a_1, a_2, \dots, a_n$ if and only if the linear system whose augmented matrix is $(**)$ is consistent.

→ function with input (finite) set of vectors output (infinite) set of vectors.
Definition. The span of a list of vectors $v_1, v_2, \dots, v_p \in \mathbb{R}^n$ is the set of all vectors $y \in \mathbb{R}^n$ that are linear combinations of $v_1, v_2, \dots, v_p$. We denote this (usually infinite) set by

$$\mathbb{R}\text{-span}\{v_1, v_2, \dots, v_p\} \quad \text{or} \quad \text{span}\{v_1, v_2, \dots, v_p\}.$$

Corollary. If $v_1, v_2, \dots, v_p \in \mathbb{R}^n$, then a vector $y \in \mathbb{R}^n$ belongs to $\mathbb{R}\text{-span}\{v_1, v_2, \dots, v_p\}$ if and only if the $n \times (p+1)$ matrix $\begin{bmatrix} v_1 & v_2 & \cdots & v_p & y \end{bmatrix}$ is the augmented matrix of a consistent linear system.

What does $\mathbb{R}\text{-span}\{v_1, v_2, \dots, v_p\}$ look like?

We can visualize the span of the 0 vector as the single point consisting of just the origin. We imagine the span of a collection of vectors that all belong to the same line through the origin as that line.

In $\mathbb{R}^2$, if the span of $v_1, v_2, \dots, v_p$ does not consist of a line, then the span is the whole plane.

To see this, imagine we have two vectors $u, v \in \mathbb{R}^2$ that are not parallel. Then we can get to any point in the plane by traveling some distance in the u direction, then some distance in the v direction. In other words, we can get any vector in $\mathbb{R}^2$ as the linear combination $au + bv$ for some scalars $a, b \in \mathbb{R}$. Draw a picture to illustrate this to yourself.

$$\left\{ u_1v_1 + u_2v_2 + \cdots + u_pv_p \mid u_1, u_2, \dots, u_p \in \mathbb{R} \right\}$$

3 Vocabulary

Keywords from today's lecture:

1. **Vector.**

A vertical list of numbers. Equivalently, a matrix with one column.

The set of all vectors with n rows is written $\mathbb{R}^n$.

Example: $\begin{bmatrix} 1 \\ 0 \\ -5.2 \\ 3 \end{bmatrix}$ or $\begin{bmatrix} 4 \end{bmatrix}$ or $\begin{bmatrix} \sqrt{2} \\ \pi \end{bmatrix}$.

2. **Scalar.**

Another word for “number” or “constant.” We can multiply scalars together, but not vectors.

Example: 5 or π or $\sqrt{2}$.

3. The **zero vector** $0 \in \mathbb{R}^n$.

The vector $\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$ with n rows all equal to zero.

4. **Linear combination** of vectors.

If $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ are vectors, then $u + v = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$.

If $c \in \mathbb{R}$ is a scalar then $cv = \begin{bmatrix} cv_1 \\ cv_2 \end{bmatrix}$.

The linear combination of vectors $v_1, v_2, \dots, v_p \in \mathbb{R}^n$ with coefficients $a_1, a_2, \dots, a_p \in \mathbb{R}$ is

$$a_1 v_1 + a_2 v_2 + \dots + a_p v_p \in \mathbb{R}^n.$$

Example: $2 \begin{bmatrix} 1 \\ 4 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \pi \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 - 0 + \pi \\ 8 - 1 + 3\pi \end{bmatrix} = \begin{bmatrix} 2 + \pi \\ 7 + 3\pi \end{bmatrix}$.

5. The **span** of a list of vectors $v_1, v_2, \dots, v_p \in \mathbb{R}^n$.

The set of all linear combinations of the vectors $v_1, v_2, \dots, v_p \in \mathbb{R}^n$.

A vector $u \in \mathbb{R}^n$ belongs to the span of $v_1, v_2, \dots, v_p \in \mathbb{R}^n$ if and only if the $n \times (p+1)$ matrix

$$A = \begin{bmatrix} v_1 & v_2 & \cdots & v_p & u \end{bmatrix}$$

is the augmented matrix of a consistent linear system.

This happens precisely when A has no pivot positions in the last column.